

INTERNSHIP PROJECT REPORT

UNEMPLOYMENT ANALYSIS IN INDIA USING PYTHON

Submitted by

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Submitted to

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Project Title

Unemployment Analysis in India Using Python

Tools Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Visual Studio Code

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ABSTRACT

Unemployment is one of the most significant socio-economic indicators that reflects the economic condition of a country. Analyzing unemployment data helps governments, researchers, and policymakers understand employment trends and formulate effective strategies for economic development.

The objective of this project is to analyze unemployment trends in India using Python-based data analysis techniques. The project utilizes a publicly available unemployment dataset containing information on unemployment rate, estimated employment, labour participation rate, region, and time period.

The dataset was imported using the Pandas library and underwent preprocessing steps such as handling missing values, removing unnecessary spaces, checking duplicate records, and converting date values into a suitable format for analysis. Descriptive statistical analysis was performed to summarize the characteristics of the dataset.

Several visualization techniques were implemented using Matplotlib and Seaborn, including a histogram to study the distribution of unemployment rates, a region-wise bar chart to compare unemployment across different states, a monthly trend analysis using a line graph, and a correlation heatmap to understand relationships among numerical variables.

The analysis revealed noticeable regional differences in unemployment rates. The average unemployment rate in the cleaned dataset was approximately **11.78%**. **Tripura** recorded the highest average unemployment rate, while **Meghalaya** showed the lowest average unemployment rate. The visualizations also highlighted fluctuations in unemployment over time, indicating the influence of economic conditions on employment patterns.

1. INTRODUCTION

Unemployment is a major economic and social issue that affects the growth and development of every nation. It refers to the situation where individuals who are willing and able to work are unable to find suitable employment. A high unemployment rate not only reduces the standard of living but also affects economic productivity, national income, and social stability.

India, being one of the world's largest developing countries, faces several employment-related challenges due to rapid population growth, changing economic conditions, technological advancements, and fluctuations in industrial development. Monitoring unemployment trends is therefore essential for understanding the country's economic condition and for developing policies that promote sustainable employment.

With the rapid growth of data science, large volumes of employment data can now be analyzed efficiently using programming languages such as Python. Data analysis techniques help transform raw datasets into meaningful information that supports evidence-based decision-making.

This project focuses on analyzing unemployment data in India using Python. The dataset was cleaned, processed, and analyzed using various data analysis libraries including Pandas, Matplotlib, and Seaborn. Statistical analysis and visualization techniques were employed to identify unemployment patterns, regional differences, and time-based trends. The project demonstrates the practical application of data science techniques in solving real-world socio-economic problems.

2. OBJECTIVES

The objectives of this project are:

- To understand the structure and characteristics of the unemployment dataset.
- To perform data cleaning and preprocessing using Python.
- To identify missing values and remove inconsistencies from the dataset.
- To analyze unemployment rates across different regions of India.
- To visualize unemployment trends using statistical graphs.
- To study the relationship between unemployment rate, employment, and labour participation rate.
- To identify regions with the highest and lowest unemployment rates.
- To generate meaningful insights using exploratory data analysis.

- To strengthen practical skills in Python programming and data visualization.
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3. PROBLEM STATEMENT

Unemployment remains one of the most significant socio-economic challenges in India. Variations in employment opportunities across different regions make it difficult to understand the overall employment scenario without proper data analysis.

Traditional methods of analyzing employment statistics are time-consuming and often fail to provide clear visual insights. Therefore, there is a need for an efficient analytical approach that can process large datasets, identify important trends, and present the results in a simple and understandable manner.

This project aims to address this problem by using Python-based data analysis techniques to explore unemployment data, generate informative visualizations, and extract useful insights that can support future policy decisions and research.

4. SCOPE OF THE PROJECT

The scope of this project includes the analysis of unemployment data collected from different regions of India. The project mainly focuses on exploratory data analysis rather than predictive modeling.

The scope includes:

- Importing and understanding the dataset.
- Cleaning and preprocessing raw data.
- Performing descriptive statistical analysis.
- Visualizing unemployment trends using charts.
- Comparing unemployment rates among different regions.
- Studying relationships between numerical variables.
- Identifying important observations from the dataset.

Although this project does not include machine learning algorithms, it provides a strong foundation for future predictive analysis and employment forecasting.

5. TOOLS AND TECHNOLOGIES USED

The following tools and technologies were used during the development of this project:

Python

Python is a powerful and widely used programming language for data analysis, scientific computing, and machine learning. It provides numerous libraries that simplify data manipulation and visualization.

Pandas

Pandas is an open-source Python library used for reading, cleaning, organizing, and analyzing structured datasets. It allows efficient handling of large datasets using DataFrames.

Matplotlib

Matplotlib is a visualization library used to create high-quality graphs such as histograms, line charts, and bar charts. It helps present analytical results in an understandable format.

Seaborn

Seaborn is built on top of Matplotlib and provides advanced statistical visualizations with attractive designs. It was used to generate the bar chart and correlation heatmap.

Visual Studio Code (VS Code)

Visual Studio Code is a lightweight and powerful source code editor used for writing, debugging, and executing Python programs.

6. SOFTWARE AND HARDWARE REQUIREMENTS

Software Requirements

Software	Version
Operating System	Windows 11
Python	3.14.5
Visual Studio Code	Latest Version
Pandas	3.0.3
Matplotlib	3.11.0
Seaborn	0.13.2

Hardware Requirements

Component Specification

Processor	Intel Core i3 or above
RAM	Minimum 4 GB (8 GB Recommended)
Storage	1 GB Free Space
Display	1366 × 768 Resolution or Higher

7. DATASET DESCRIPTION

The dataset used in this project contains unemployment statistics collected from different regions of India. It provides information regarding unemployment rate, employment count, labour participation rate, and geographical area over multiple time periods.

Initially, the dataset contained **768 records** with **7 attributes**. During preprocessing, **28 records containing missing values** were removed, resulting in a cleaned dataset consisting of **740 records**.

The dataset contains the following attributes:

Attribute	Description
Region	Name of the state or union territory
Date	Date of observation
Frequency	Frequency of data collection (Monthly)
Estimated Unemployment Rate (%)	Percentage of unemployed individuals
Estimated Employed	Number of employed individuals
Estimated Labour Participation Rate (%)	Percentage of labour force participation
Area	Rural or Urban classification

The dataset serves as a valuable source for understanding employment patterns across different regions and identifying economic trends.

8. PROJECT METHODOLOGY

The project was completed through a systematic sequence of steps to ensure accurate analysis and meaningful results.

Step 1: Dataset Collection

The unemployment dataset was collected and stored in CSV format. The dataset was placed inside the project directory for easy access through Python.

Step 2: Data Loading

The dataset was imported into Python using the Pandas library. Initial exploration was performed using functions such as `head()`, `shape()`, `columns`, and `info()` to understand the structure of the dataset.

Step 3: Data Cleaning

The preprocessing stage involved removing missing values, correcting column names, stripping unnecessary spaces, converting date values into datetime format, and checking for duplicate records.

Step 4: Exploratory Data Analysis

Descriptive statistical analysis was performed using the `describe()` function. Various visualizations including histogram, region-wise bar chart, monthly trend graph, and correlation heatmap were generated to understand unemployment patterns.

Step 5: Insight Generation

The cleaned data was analyzed to identify average unemployment rate, highest and lowest unemployment values, and regions with maximum and minimum unemployment. These insights were summarized to support the conclusions of the project.

9. PROJECT IMPLEMENTATION

This section describes the practical implementation of the project using Python and various data analysis libraries. The dataset was loaded, cleaned, analyzed, and visualized to understand unemployment patterns in India.

9.1 Importing Required Libraries

The first step was to import the necessary Python libraries required for data analysis and visualization.

The following libraries were used:

- **Pandas** for data manipulation and analysis.

- **Matplotlib** for creating visualizations.
- **Seaborn** for advanced statistical plots.

Code Used

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

9.2 Loading the Dataset

The unemployment dataset was loaded into a Pandas DataFrame using the `read_csv()` function. The first few records were displayed to verify successful loading.

Code Used

```
df = pd.read_csv("Unemployment in India.csv")
print(df.head())
```

The output confirmed that the dataset was imported correctly and contained information about unemployment rate, employment figures, labour participation rate, and region.

9.3 Exploring the Dataset

Before performing analysis, the structure of the dataset was examined using various functions.

Code Used

```
print(df.shape)
print(df.columns)
print(df.info())
```

This step helped determine:

- Number of rows and columns.
- Available attributes.
- Data types.
- Presence of missing values.

Output Obtained

- Total Records: **768**

- Total Columns: **7**
-

9.4 Data Cleaning and Preprocessing

Real-world datasets often contain inconsistencies and missing values. Therefore, preprocessing was performed before analysis.

The following operations were carried out:

Cleaning Column Names

```
df.columns = df.columns.str.strip()
```

Checking Missing Values

```
print(df.isnull().sum())
```

Removing Missing Values

```
df = df.dropna()
```

Checking Duplicate Records

```
print(df.duplicated().sum())
```

Results

- Missing records identified: **28**
 - Missing records removed successfully.
 - Duplicate records found: **0**
 - Final dataset size: **740 rows**
-

9.5 Statistical Analysis

Descriptive statistical analysis was performed using the `describe()` function.

Code Used

```
print(df.describe())
```

This provided important statistical measures such as:

- Mean
- Standard Deviation
- Minimum Value

- Maximum Value
- Quartiles

Key Findings

Statistic	Value
Average Unemployment Rate	11.78%
Maximum Unemployment Rate	76.74%
Minimum Unemployment Rate	0.00%

9.6 Histogram of Unemployment Rate

A histogram was created to understand the distribution of unemployment rates.

Code Used

```
plt.hist(df["Estimated Unemployment Rate (%)"], bins=20)
plt.savefig("Histogram.png")
```

The histogram shows how unemployment rates are distributed across different observations.

Observation

Most unemployment values are concentrated at lower percentages, while a smaller number of observations exhibit very high unemployment rates.

16.7 Region-wise Unemployment Analysis

A bar chart was created to compare unemployment rates among different regions.

Code Used

```
sns.barplot(
    x="Region",
    y="Estimated Unemployment Rate (%)",
    data=df
)
```

Observation

The graph reveals significant differences in unemployment levels among regions, indicating varying economic conditions.

9.8 Monthly Unemployment Trend Analysis

Time-series analysis was performed to observe unemployment variations over time.

Code Used

```
df["Date"] = df["Date"].str.strip()
df["Date"] = pd.to_datetime(df["Date"])

sns.lineplot(
    x="Date",
    y="Estimated Unemployment Rate (%)",
    data=df
)
```

Observation

The trend graph indicates fluctuations in unemployment rates over different months, reflecting changes in economic activity.

9.9 Correlation Analysis

A heatmap was generated to identify relationships between numerical variables.

Code Used

```
sns.heatmap(
    df.corr(numeric_only=True),
    annot=True
)
```

Observation

The heatmap provides a visual representation of the correlation between unemployment rate, employment count, and labour participation rate.

9.10 Key Insights Generated

After completing the analysis, important insights were extracted from the dataset.

Output Obtained

Metric	Value
Average Unemployment Rate	11.78%
Highest Unemployment Rate	76.74%
Lowest Unemployment Rate	0.00%
Highest Unemployment Region	Tripura (28.35%)
Lowest Unemployment Region	Meghalaya (4.79%)

Interpretation

The analysis indicates that unemployment varies significantly across regions. Tripura exhibited the highest average unemployment rate, whereas Meghalaya showed the lowest unemployment rate. The dataset also demonstrates periods of sharp increases in unemployment, suggesting the influence of economic and social factors on employment conditions.

10. RESULTS AND DISCUSSION

The unemployment dataset was successfully analyzed using Python-based data science techniques. Data cleaning improved dataset quality by removing incomplete records and standardizing column names. Statistical analysis provided a summary of employment-related variables, while visualizations offered a clear understanding of unemployment patterns.

The histogram revealed that most observations fall within lower unemployment ranges, whereas a few observations exhibited extremely high unemployment values. Region-wise analysis showed substantial variation among states, highlighting economic disparities. The monthly trend graph demonstrated fluctuations in unemployment over time, while the correlation heatmap provided insight into relationships among numerical variables.

Overall, the analysis successfully transformed raw data into meaningful information that can support economic understanding and future research.

11. KEY FINDINGS

The analysis of the unemployment dataset revealed several important observations regarding employment trends across different regions of India. The major findings are summarized below:

1. The cleaned dataset consisted of **740 records** and **7 attributes** after removing missing values.
2. The **average unemployment rate** observed in the dataset was **11.78%**, indicating moderate unemployment across the country during the study period.
3. The **highest unemployment rate** recorded was **76.74%**, suggesting that certain regions experienced severe employment challenges during specific periods.
4. The **lowest unemployment rate** was **0.00%**, indicating periods or regions with negligible unemployment.
5. Among all the regions analyzed, **Tripura** recorded the highest average unemployment rate (**28.35%**), highlighting a comparatively weaker employment scenario.
6. **Meghalaya** recorded the lowest average unemployment rate (**4.79%**), indicating relatively better employment conditions.
7. The histogram demonstrated that most unemployment observations were concentrated within lower unemployment ranges, while a smaller number of observations showed exceptionally high unemployment values.
8. The region-wise bar chart clearly illustrated significant differences in unemployment among Indian states and union territories.
9. The monthly trend graph showed noticeable fluctuations in unemployment over time, indicating that employment conditions vary due to economic, seasonal, and social factors.
10. The correlation heatmap provided an understanding of the relationships between unemployment rate, employment count, and labour participation rate, enabling better interpretation of the dataset.

These findings demonstrate how data analysis techniques can transform raw data into meaningful information that supports informed decision-making.

12. CONCLUSION

The **Unemployment Analysis in India Using Python** project was successfully completed using various data science techniques. The project involved importing a

real-world unemployment dataset, cleaning and preprocessing the data, performing descriptive statistical analysis, generating informative visualizations, and extracting valuable insights.

Python libraries such as **Pandas**, **Matplotlib**, and **Seaborn** proved highly effective in handling and analyzing the dataset. The preprocessing stage ensured improved data quality by removing missing values and correcting formatting inconsistencies. Statistical analysis provided a clear understanding of the overall characteristics of the dataset, while graphical visualizations simplified the interpretation of unemployment trends.

The analysis highlighted considerable regional variation in unemployment across India. Some regions experienced consistently higher unemployment rates, whereas others maintained relatively stable employment conditions. The time-series analysis further demonstrated that unemployment fluctuates over different months, emphasizing the influence of changing economic conditions.

This project enhanced practical knowledge of data preprocessing, exploratory data analysis, visualization, and interpretation of real-world datasets. It also strengthened programming skills in Python and provided valuable experience in solving real-life analytical problems. Overall, the project successfully achieved its objectives and demonstrated the importance of data science in understanding socio-economic issues.

13. FUTURE SCOPE

Although the current project focuses on exploratory data analysis, it can be extended in several ways to enhance its practical applications.

Future improvements include:

- Developing machine learning models to predict future unemployment rates.
- Creating interactive dashboards using **Power BI** or **Tableau** for better visualization.
- Integrating additional economic indicators such as GDP, inflation, literacy rate, and industrial growth for more comprehensive analysis.
- Performing state-wise forecasting of unemployment trends using time-series models.
- Deploying the project as a web application using **Streamlit** or **Flask** for interactive analysis.
- Comparing unemployment trends across different countries to understand global employment patterns.

These enhancements would provide deeper insights and improve the usefulness of the project for policymakers, researchers, and organizations.

14. REFERENCES

The following references were used during the completion of this project:

1. McKinney, W. (2022). *Python for Data Analysis* (3rd Edition). O'Reilly Media.
2. VanderPlas, J. (2016). *Python Data Science Handbook*. O'Reilly Media.
3. Wes McKinney. *Pandas Documentation*.
4. Matplotlib Development Team. *Matplotlib Documentation*.
5. Seaborn Development Team. *Seaborn Statistical Data Visualization Documentation*.
6. Python Software Foundation. *Python Programming Language Documentation*.
7. Government employment and unemployment statistical datasets used for educational purposes.

APPENDIX

Project Output Summary

Parameter	Result
Programming Language	Python
IDE	Visual Studio Code
Dataset Records	768
Cleaned Records	740
Missing Values Removed	28
Duplicate Records	0
Average Unemployment Rate	11.78%
Highest Unemployment Rate	76.74%
Lowest Unemployment Rate	0.00%

Parameter	Result
Highest Unemployment Region	Tripura
Lowest Unemployment Region	Meghalaya

List of Generated Outputs

The following outputs were successfully generated during the project:

- Dataset Preview
 - Dataset Information
 - Data Cleaning Summary
 - Statistical Summary
 - Histogram of Estimated Unemployment Rate
 - Region-wise Unemployment Bar Chart
 - Monthly Unemployment Trend Line Graph
 - Correlation Heatmap
 - Key Insights Summary
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INTERNSHIP LEARNING OUTCOMES

Through this internship project, the following technical skills were acquired and strengthened:

- Python Programming
- Data Import and Data Cleaning
- Data Manipulation using Pandas
- Exploratory Data Analysis (EDA)
- Statistical Analysis
- Data Visualization using Matplotlib and Seaborn
- Interpretation of Analytical Results
- Preparation of Professional Technical Reports
- Practical Understanding of Real-World Socio-economic Data

PROJECT COMPLETION STATEMENT

This internship project entitled "**Unemployment Analysis in India Using Python**" has been successfully completed in accordance with the objectives specified at the beginning of the study. The project demonstrates the practical application of Python-based data analysis techniques for understanding unemployment trends through effective preprocessing, visualization, and interpretation of real-world data.

The successful completion of this project has enhanced both technical and analytical skills, providing a strong foundation for future work in **Data Science**, **Bioinformatics**, **Machine Learning**, and **Business Analytics**.